

3(a)(i)	<u>metaphase</u> , <u>II</u> / <u>2</u> ;	1
3(a)(ii)	<i>any two from</i> 1 spindle (fibres) / microtubules, contract / shorten ; 2 <u>chromosome(s)</u> , move / pulled, to pole(s) / centriole(s) ; 3 <u>centromeres</u> , move / lead / pulled, first ; 4 homologous chromosomes <u>separate</u> or pair (of chromosomes) / bivalent, pulled apart ;	2

Question	Answer	Marks
3(a)(iii)	<i>any two from</i> 1 <u>non-sister chromatids</u> ; 2 swap / exchange, alleles / sections of DNA ; 3 <u>chiasmata</u> ; 4 (giving) new / different, <u>allele</u> combinations or recombination ;	2
3(b)(i)	<i>any two from</i> 1 test (leaves / population / group) takes, longer / higher time or test group degrades mesotrione more slowly ; 2 so test group, less resistant / fewer have resistance ; 3 test standard deviation is, larger / higher, so (population) more variable ;	2
3(b)(ii)	<i>any two from</i> 1 can, adapt / evolve / survive long term ; 2 resistant / mutant, plants / hemp, have selective advantage ; 3 in areas where / when, mesotrione is used ; 4 resistant / mutant, plants survive and reproduce ;	2
3(b)(iii)	<i>any one from</i> 1 lets farmer choose, different / correct / effective / best, herbicide ; 2 less money wasted due to, wrong herbicide / reduced crop yield ;	1
3(c)(i)	<i>numerator working</i> 17.4 ; <i>denominator working</i> $\sqrt{3.76 + 0.91}$ OR $\sqrt{4.67}$ OR 2.16 ;	2
3(c)(ii)	<i>any two from</i> 1 <u>reject</u> because, t / calculated value / it, is, higher / greater / more, than, 2.23 / <u>critical</u> value ; 2 probability of result by chance is less than, 0.05 / 5% or $p < 0.05$; 3 there is a significant difference between the, means / populations / test and control ;	2